ADVANCED COMPUTER LAB · MILESTONE 1 · ANALYTICAL REPORT

Formula 1 Podium Prediction

Data-engineering answers · Selected features and pre-race validity · Known limitations

Team 106		Links	
John Magdy Fouad Soliman	58-22411	Kaggle	kaggle.com/code/johnsolimann/project-acl-2026
Abdelrahman Mohamed Abdelrahman Omar Elaswar	58-4203	GitHub	github.com/Johnn-19/Advanced-Computer-Labs-2026
Omar Ehab Mohamed Salama Nasr	58-0949	Dataset	Kaggle / Ergast F1 World Championship 1950–2024
Mohamed Hany Elghamry Youssef	58-1835	Date	18 October 2026

Context

For every driver in every race we predict whether the driver finishes on the podium (`positionOrder ≤ 3`), using only information available **after qualifying and before the race starts**. The 14 raw tables were cleaned into a modeling table with exactly one row per `(raceId, driverId)`: 24,551 real race starts, with the Indianapolis 500, non-starters and shared-drive duplicates removed, and a podium rate of 13.6%. The split is by season: train 1950–2019, validation 2020–2021, test 2022–2024. This report covers the three parts the milestone asks for. All numbers and figures come from the submitted run-all notebook.

All three questions need **multi-hop joins**, because `results` alone has no season, circuit, country, nationality, team name or retirement reason. The backbone is left-joined once to `races`, `circuits`, `drivers`, `constructors`, `status` and `qualifying`. An assertion confirms that no row is added or lost (24,551 rows, grain intact). After the join only the qualifying position is ever missing (57.6%), because qualifying is recorded only from 1994 and is patchy in 1996–2002. The questions use post-race columns (`positionOrder`, `status`) only as the *outcome being studied*; they are never model features.

1. Data-Engineering Questions

Q1 — Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Method. *Front row by grid* means `grid` $\in \{1, 2\}$, where the car actually started after penalties. *Front row by qualifying* means qualifying position $\in \{1, 2\}$, before penalties. Conversion rate = podiums ÷ front-row starts per circuit. We compare 2003–2013 (pre-hybrid) with 2014–2024 (hybrid and after). Both windows are 11 seasons long and have essentially complete qualifying data, so the two definitions can be compared on the same races. A circuit is ranked only with ≥ 10 front-row starts in a window, and every rate has a 95% Wilson interval.

Table 1 — Top-10 circuits by front-row (grid) conversion

#	2003–2013	Conversion	2014–2024	Conversion
1	Circuit de Nevers Magny-Cours	91.7% (11/12)	Circuit de Monaco	89.5% (17/19)
2	Istanbul Park	78.6% (11/14)	Suzuka Circuit	88.9% (16/18)
3	Autodromo Nazionale di Monza	77.3% (17/22)	Yas Marina Circuit	86.4% (19/22)
4	Circuit de Monaco	77.3% (17/22)	Circuit de Barcelona-Catalunya	81.8% (18/22)
5	Silverstone Circuit	72.7% (16/22)	Circuit de Spa-Francorchamps	81.8% (18/22)
6	Circuit de Spa-Francorchamps	72.2% (13/18)	Sochi Autodrom	81.2% (13/16)
7	Suzuka Circuit	72.2% (13/18)	Circuit of the Americas	80.0% (16/20)
8	Hockenheimring	71.4% (10/14)	Autódromo José Carlos Pace	78.9% (15/19)
9	Circuit de Barcelona-Catalunya	68.2% (15/22)	Shanghai International Circuit	78.6% (11/14)
10	Bahrain International Circuit	66.7% (12/18)	Circuit Gilles Villeneuve	77.8% (14/18)

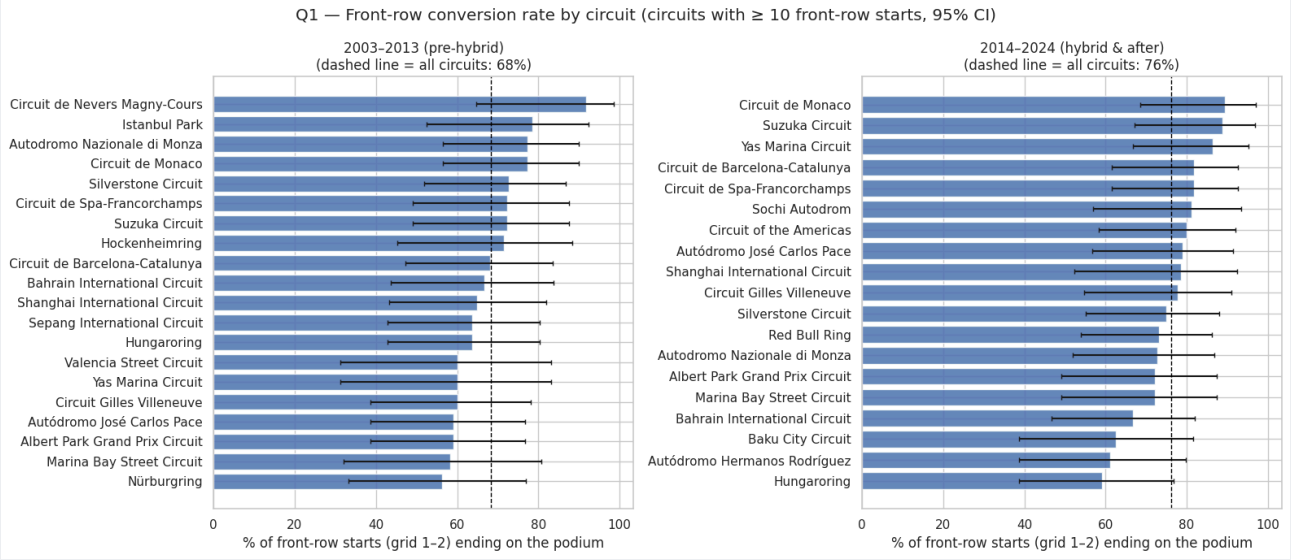


Figure 1. Q1: front-row (grid 1–2) conversion by circuit with 95% Wilson intervals; dashed line = all circuits (68% pre-2014, 76% post-2014). With only ≈ 10–22 front-row starts per circuit and window, neighbouring circuits are not statistically different.

Ranking change. For the 14 circuits ranked in both windows, Spearman's ρ = 0.36 (p = 0.21) with the grid definition and 0.54 (p = 0.047) with the qualifying definition. Monza fell from joint 1st to 10th, and Silverstone, Bahrain and the Hungaroring also dropped. Yas Marina (10th → 3rd) and Interlagos (12th → 6th) climbed. Monaco stayed 1st (overtaking is hardest there), and Suzuka moved from 4th to 2nd.

Grid vs qualifying. 815 drivers both qualified and started on the front row; 36 qualified there but started further back (penalties), and 36 were promoted onto it. Within each window the two definitions give almost the same ranking (ρ = 0.94 / 0.90). The outcome differs sharply, though: penalised front-row qualifiers podiumed only 30.6% of the time, versus 63.9% for promoted drivers.



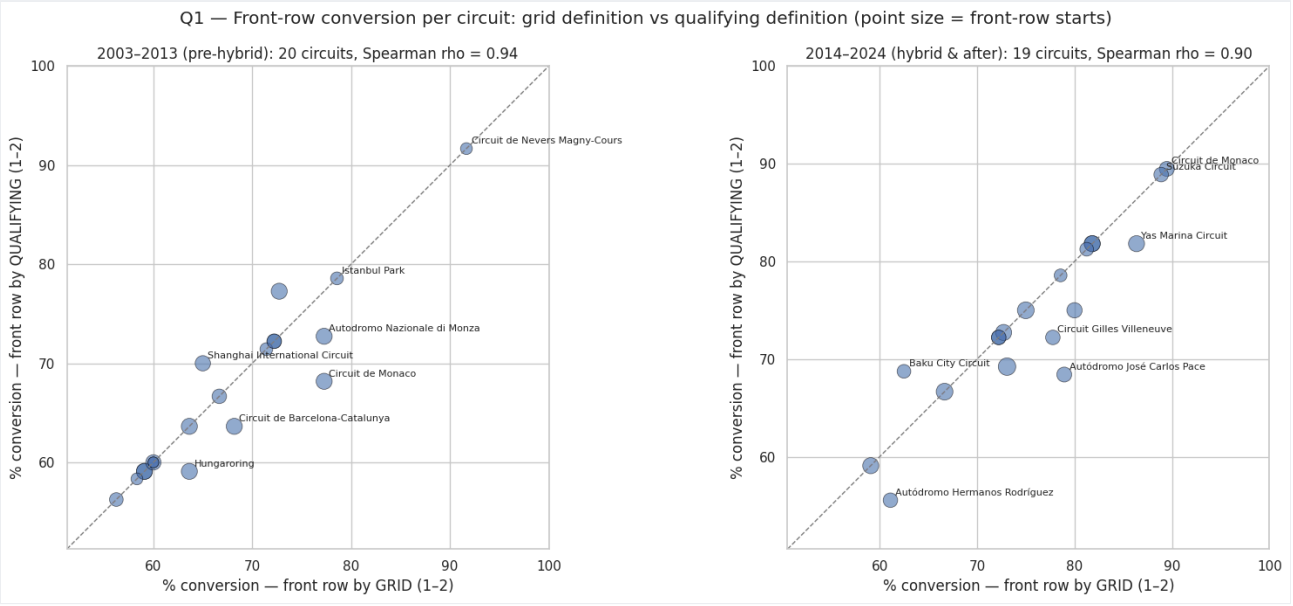


Figure 2. Top: circuit rank before vs after 2014 (red = moved at least a third of the table). Bottom: per-circuit conversion under the grid vs qualifying definition (point size = front-row starts).

Answer to Q1

In **2014–2024** the best converters are **Monaco (89.5%), Suzuka (88.9%) and Yas Marina (86.4%)**. In **2003–2013** they were **Magny-Cours (91.7%, only 6 races), Istanbul Park (78.6%), and Monza and Monaco (77.3%)**. The ranking **partly changes** after 2014: the top (Monaco, Suzuka, Spa) is fairly stable while the middle and bottom reshuffle. Overall conversion rises from 68.3% to 76.2%. **Where a car starts matters more than where it qualified.**

Grain: one front-row start (driver × race), aggregated to circuit × window; circuits by `circuitId`. **Missingness:** qualifying is present for 99.8% of 2003–2024 rows and for every grid front-row starter; circuits with < 10 front-row starts are not ranked. A full-history check (1950–2013) mostly measures early-era unreliability (56% conversion, ρ ≈ −0.08). **Cutoff:** groups are defined by grid and qualifying position, both known before the race.

Q2 — Do drivers podium more often at home, after controlling for starting position?

Method. A race is a *home race* when the circuit's country is one of the driver's nationality countries. We use a documented mapping of all 42 nationality adjectives to circuit countries (dual nationalities count in both countries; "United States" = "USA"). This gives 1,929 home starts (7.9%). Without control, home drivers podium *less* (11.7% vs 13.9%), but they also start further back (median grid 13 vs 11). We therefore compare drivers **within grid buckets** (1, 2–3, 4–6, 7–10, 11–15, 16+) and pool them with the Cochran–Mantel–Haenszel (CMH) odds ratio and with logistic regression `podium ~ is_home + grid + decade`. The 72 pit-lane starters are left out.

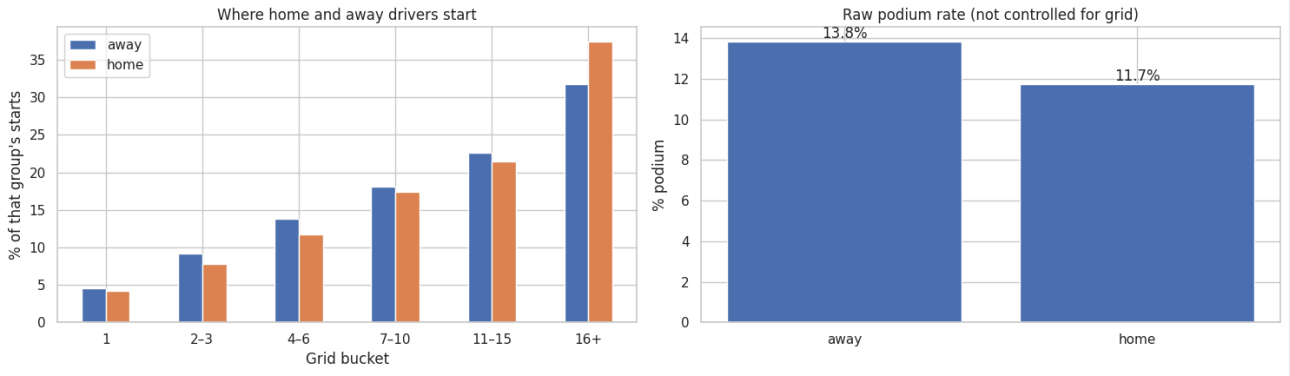


Figure 3. Q2 without any control: where home and away drivers start (left) and their raw podium rate (right). Home drivers start further back, which explains their lower raw podium rate.

Table 2 — Home effect before and after controlling for starting position

Method	Odds ratio	95% CI	p-value	Rows
Raw (no control)	0.83	0.72–0.95	0.009	24,479
CMH (within grid buckets)	0.93	0.78–1.10	0.41	24,479
Logit: grid bucket + decade	0.92	0.78–1.10	0.37	24,479
Logit: exact grid (1–20) + decade	0.94	0.79–1.12	0.49	21,733
Logit: grid bucket + decade, 1980+	0.92	0.73–1.15	0.45	17,700

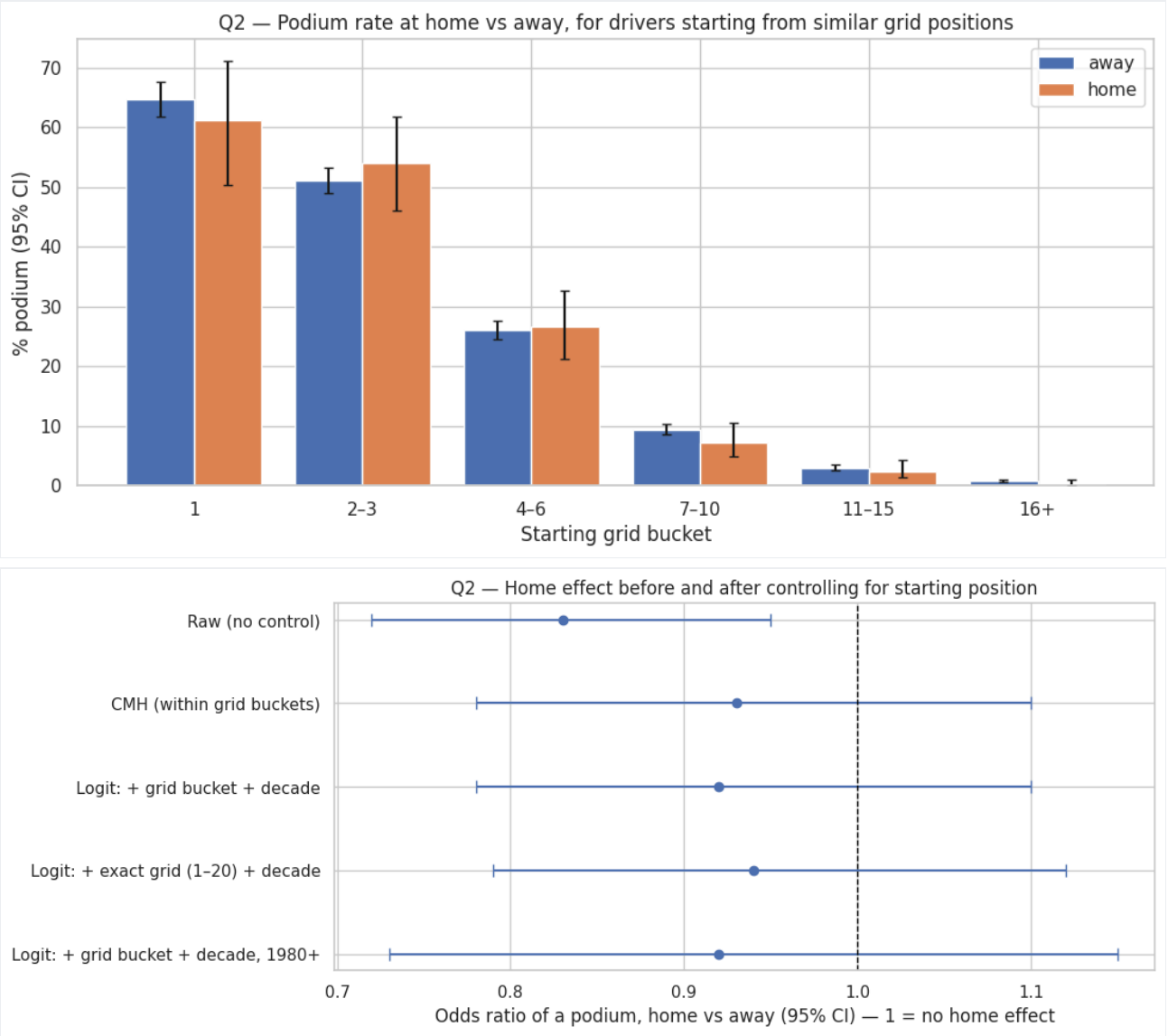


Figure 4. Top: podium rate at home vs away for drivers starting from similar grid positions (95% CI). The difference changes sign between buckets and the intervals always overlap. Bottom: odds ratio of a podium, home vs away; only the uncontrolled estimate excludes 1.

> **Answer to Q2 — No**
>
> After controlling for starting position, drivers do **not** podium more often at home. The controlled odds ratio is 0.92–0.94 with every method (CMH 0.93, 95% CI 0.78–1.10, p = 0.41), and the Breslow–Day test (p = 0.46) supports pooling the buckets. The raw "home disadvantage" exists only because home drivers start further back.
>
> **Grain:** one row per driver race start. **Missingness:** nationality and circuit country are never missing; 14 nationalities never had a home race; home starts are ≈ 8% of rows, so small effects cannot be detected. **Cutoff:** `is_home` is known before the race. The trained model also gives it ≈ 0 importance. This is an association, not a causal estimate.

Q3 — Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Method. A mechanical retirement is a race start that ended because a car component failed. All 139 status values are mapped explicitly to one category (asserted): 63 mechanical (Engine, Gearbox, Hydraulics, Brakes, Power Unit, …), 34 finished, 16 ambiguous (*Retired*, *Puncture*, wing/wheel failures, …), and accident/driver, health, regulation and other. Ambiguous statuses are excluded from "mechanical" in the main rule and included in a sensitivity check. Rate = mechanical retirements ÷ race starts per `constructorId` and era, for teams with ≥ 40 starts in the era.

Table 3 — Highest mechanical-retirement rates per constructor

#	2014–2021	Rate (mech / starts)	2022–2024	Rate (mech / starts)
1	**Lotus F1**	**17.3% (13/75)**	**Alfa Romeo**	**12.5% (11/88)**
2	**Toro Rosso**	**16.1% (39/242)**	**Alpine F1 Team**	**10.4% (14/135)**
3	**Renault**	**13.1% (26/199)**	**AlphaTauri**	**8.0% (7/88)**
4	McLaren	12.7% (40/315)	Williams	6.0% (8/134)
5	Red Bull	10.7% (34/318)	Ferrari	5.2% (7/134)
…	most reliable: Mercedes 3.8%, Ferrari 4.4%		lowest: McLaren, Red Bull 3.7%; Sauber 2.1% (48 starts)	
	All teams	**8.9%** (287/3,243)	**All teams**	**5.7%** (77/1,351)

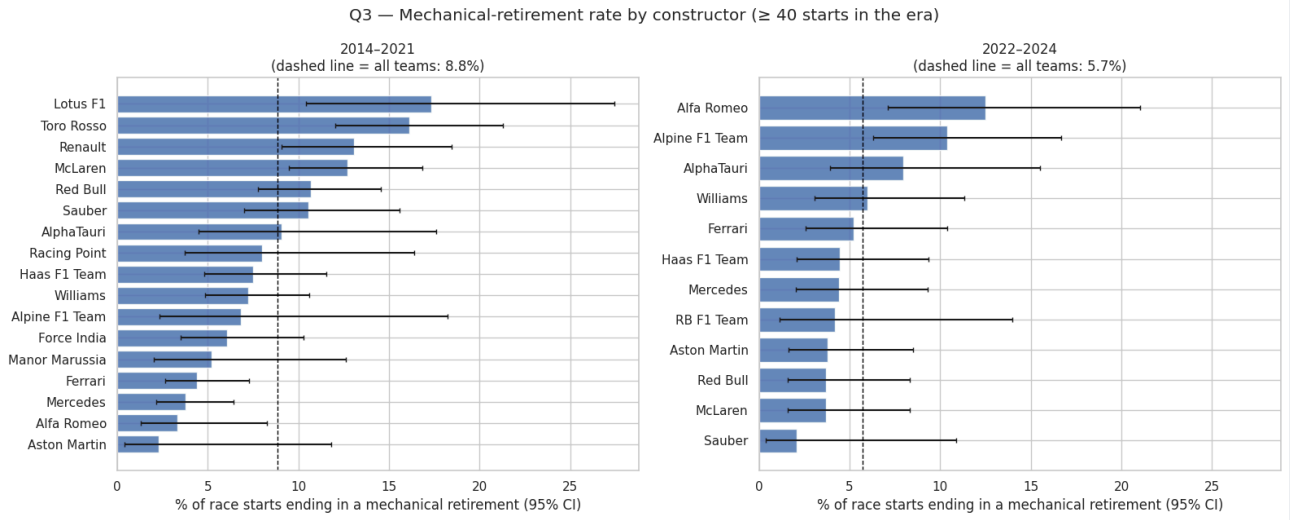


Figure 5. Q3: mechanical-retirement rate by constructor with 95% Wilson CI (dashed = all teams). The 2022–2024 counts are small (1–14 retirements per team), so below the top two the intervals overlap.

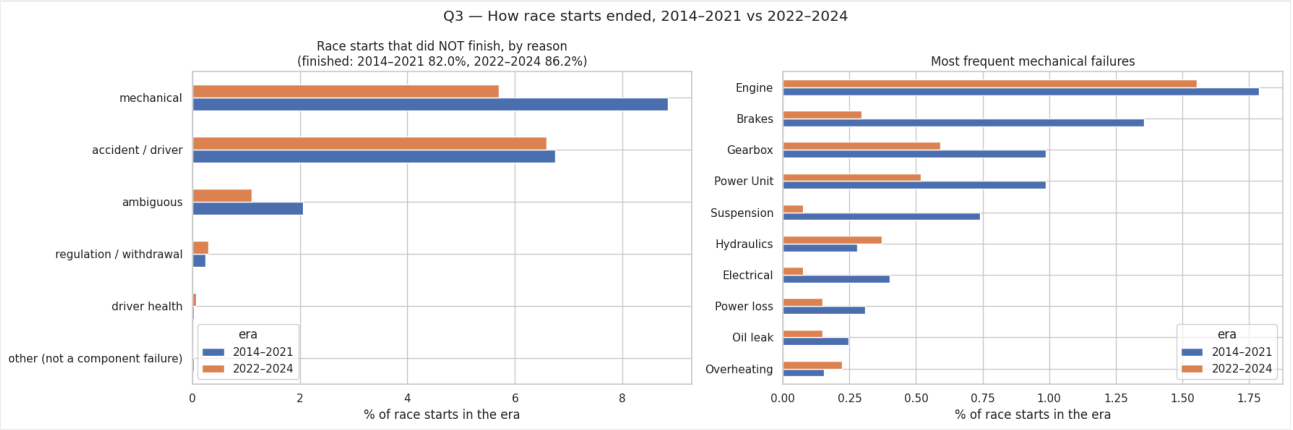


Figure 6. Q3: how race starts ended in each era, and the most frequent component failures (both as % of starts in the era). The finish rate rose from 82.0% to 86.2%, almost entirely through fewer mechanical failures; accidents stayed at ≈ 6.6–6.8%.

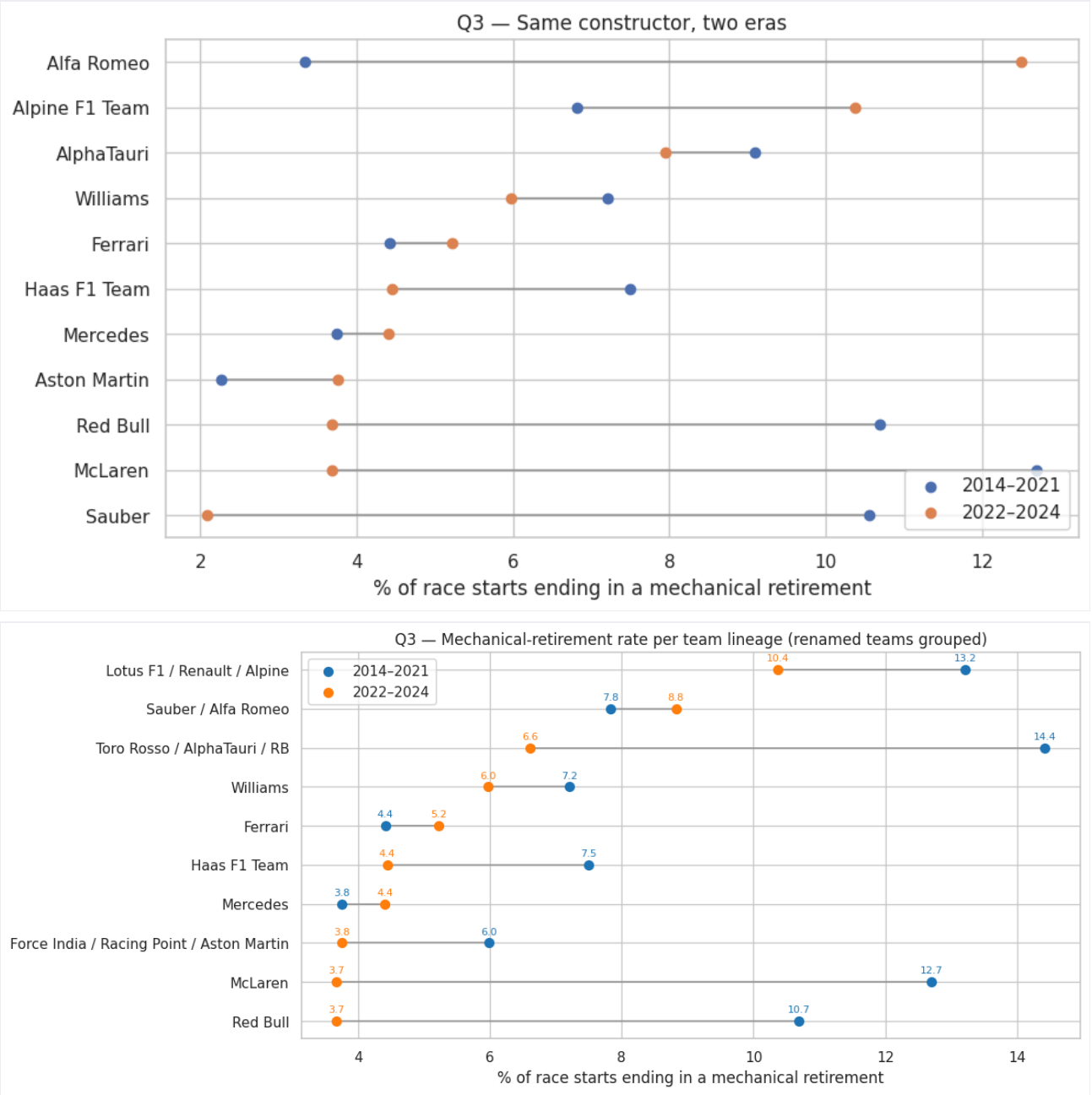


Figure 7. Top: the same constructorId in both eras. McLaren (12.7% → 3.7%) and Red Bull (10.7% → 3.7%) became much more reliable, while Alfa Romeo went from 3.3% to 12.5%. Bottom: robustness check with renamed teams grouped into lineages (e.g. Toro Rosso → AlphaTauri → RB).

Answer to Q3

2014–2021: Lotus F1 (17.3%), **Toro Rosso (16.1%)**, **Renault (13.1%)** and **McLaren (12.7%**, mostly its 2015–17 Honda seasons). **2022–2024: Alfa Romeo (12.5%)** and **Alpine (10.4%)**, then AlphaTauri (8.0%). Reliability improved overall (8.9% → 5.7% of starts), and McLaren and Red Bull improved most. The 2014–2021 ranking is robust to lineage grouping and to counting ambiguous statuses (ρ = 0.97). For 2022–2024 only the top two are robust (ρ = 0.68).

Grain: one race start, aggregated to constructor × era. **Missingness:** `status` is never missing; non-starters are excluded from the denominator; the status mapping is a documented judgement. **Cutoff:** `status` is post-race, so it is only the outcome here. Its leakage-safe version, a team's mechanical-DNF rate over *previous* races, is a model feature.

2. Selected Features and Why Each Is Valid at the Pre-Race Cutoff

The model uses **24 features in 6 groups**. They were selected with evidence from the training seasons only and with controlled validation experiments: rolling window N = 20 and full-history training beat N = 3/5/10 and the 1980+ and 2003+ windows; circuit/constructor identity, `year` and `round` were tested and rejected. The rule for validity: a feature may only use information that exists **once qualifying is over and the grid is fixed**. Three mechanisms enforce this:

- **Past-race history only.** Form features use `shift(1)` *before* the rolling window, so the value for race *t* comes from races *t−1*, *t−2*, … and never from race *t*.
- **No teammate leak.** Constructor form aggregates both cars into one constructor-race row *before* shifting, so one teammate's current result cannot reach the other's feature.
- **Previous-round standings.** The standings tables are recorded *after* each race, so they are joined from round *r−1* of the same season. Round 1 has none.

Table 4 — Final feature set and its pre-race validity

Feature(s)	Meaning	Why it is known before the race
Grid		
`grid_pos`	starting slot; a pit-lane start (`grid = 0`) is set to the last slot	the starting grid is published before the start (after penalties)
`grid_rel`	grid_pos ÷ field size (comparable across 20- and 34-car fields)	grid and number of starters are both fixed
`pit_lane_start`	flag for a pit-lane start	part of the published grid
Qualifying		
`quali_pos`	qualifying position	qualifying ends before the race
`quali_gap_pct`	% gap of the driver's best session time to the fastest qualifier of the same race	qualifying times of the same weekend
`grid_minus_quali`	grid minus qualifying position (positive = grid penalty)	both inputs are known before the start

Feature(s)	Meaning	Why it is known before the race
has_quali	whether a qualifying record exists (coverage flag)	known from the qualifying table
Driver form (previous 20 starts)		
drv_podium_rate_20, drv_finish_pct_20, drv_finish_rate_20	podium share, mean relative finishing position, finish share over the last 20 starts	computed from **earlier races only** (shift(1) then rolling)
drv_career_podium_rate	podium share over all previous starts	earlier races only (expanding mean after shift(1))
Constructor form (previous 20 races of the team)		
con_podium_rate_20, con_finish_pct_20, con_mech_dnf_rate_20	podiums per car, mean finishing position, mechanical-DNF rate (Q3 status mapping)	earlier races only, aggregated per constructor-race first, so there is no teammate leak
Championship standings (after the previous round)		
drv_champ_pos, drv_champ_pts_share, drv_season_wins	driver rank, share of the leader's points, wins so far this season	standings after round $r-1$; asserted that a winner's count excludes the current win
con_champ_pos, con_champ_pts_share	constructor rank and share of the leader's points	standings after round $r-1$
is_season_opener	round 1 flag (no standings exist yet)	race calendar
Context		
field_size	number of cars starting the race	known once the grid is fixed
is_home	circuit in the driver's country (Q2 mapping)	static nationality and circuit data
driver_age	age on race day	date of birth and race date
career_starts	number of previous starts	count of earlier races

Never used as features: current-race positionOrder, position, positionText, points, laps, time / milliseconds, fastest-lap columns, rank, status, lap times, pit stops, constructor_results and same-race standings. These are held in a FORBIDDEN_FEATURES set that is asserted against the feature list and every inference record. Sprint results are excluded: they appear in only 18 of 1,114 races, none of them in training, and in 2021–22 the sprint result already set the grid.

How validity is verified

Six assertions stop the notebook if any leak exists:

1. A driver's or team's first race has no form value.
2. Driver form recomputed by brute force from strictly earlier races matches the feature (max difference 3e-16).
3. The same brute-force check holds for constructor form (max difference 0).
4. Changing one race's results leaves that race's features unchanged but changes later races.
5. Standings come from round $r-1$ of the same season.
6. No single feature separates podiums almost perfectly (best single-feature ROC-AUC = 0.905, for `quali_pos`).

The inference function rebuilds the features of 150 random validation/test rows identically (max difference 0) and rejects records containing post-race fields.

Evidence behind the selected features

All feature evidence was computed on the **training seasons only (≤ 2019, 22,426 rows)**, so the validation and test seasons never influenced which features were kept. Each feature is justified either by an observed relationship with the podium (hypothesis-driven) or by a controlled validation experiment kept in the notebook (empirical).

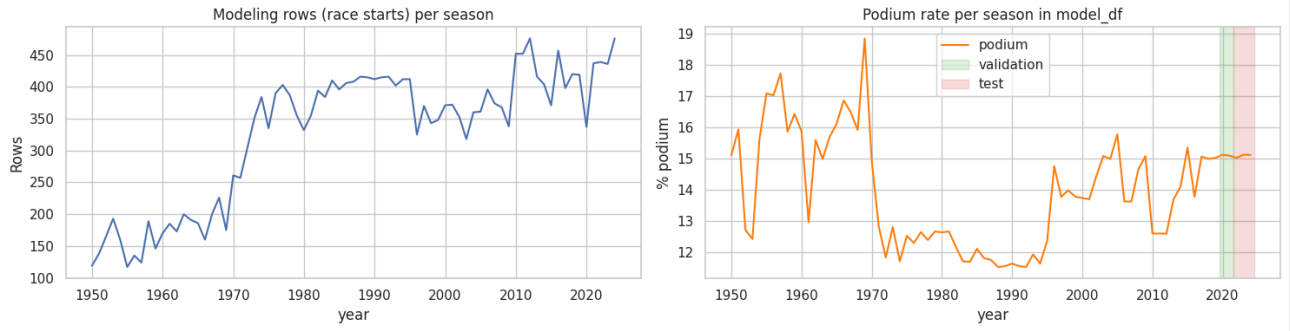


Figure 8. The modeling table: race starts per season and podium rate per season (green = validation seasons, red = test seasons). Every row is one driver in one race, unique on (raceId, driverId).

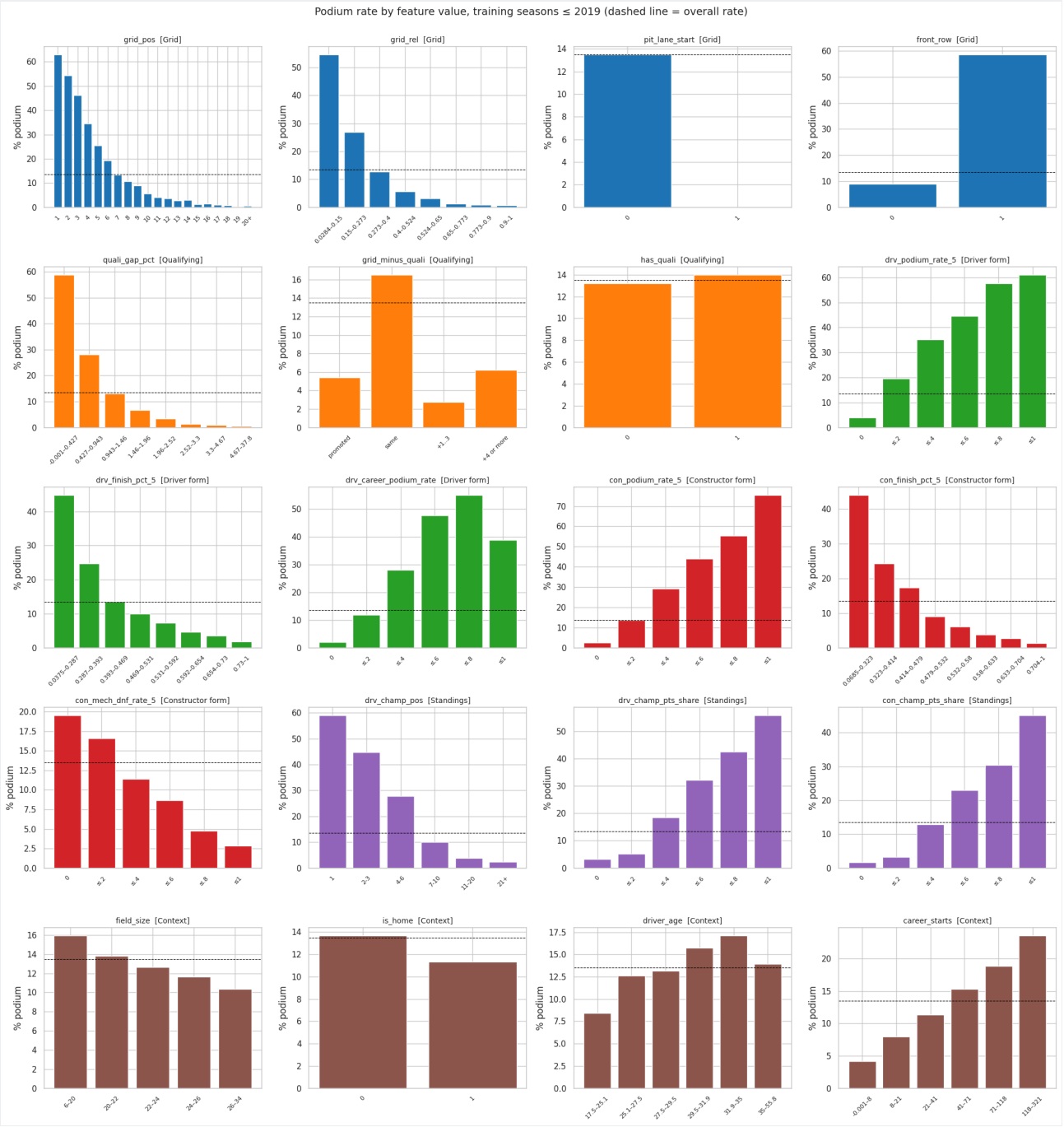


Figure 9. Podium rate across the values of each candidate feature (colour = feature group; dashed line = overall rate 13.5%). Grid: ≈ 63% podiums from pole, under 5% beyond P10. Qualifying: 59% within 0.4% of pole, under 1% beyond 3.3%. Driver form: 4% → 61%. Constructor form: 3% → 75%, and fewer podiums with more recent mechanical DNFs (20% → 3%). Standings: the championship leader podiums ≈ 59% of the time. Context features are weak; `is_home` shows no effect (as found in Q2).

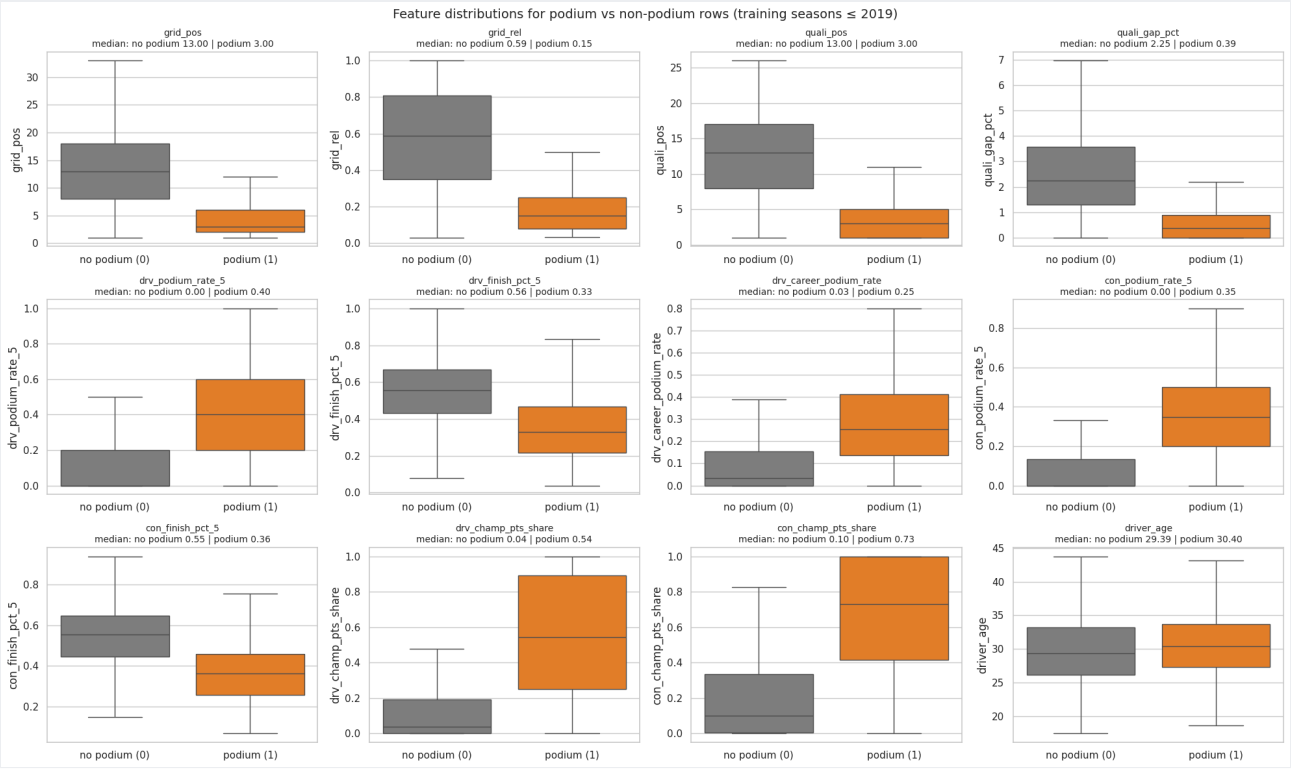


Figure 10. Distributions of the numeric candidate features for podium vs non-podium rows (box = middle 50%, line = median). Podium finishers start further forward, qualify closer to pole and come from drivers and teams in better recent form.

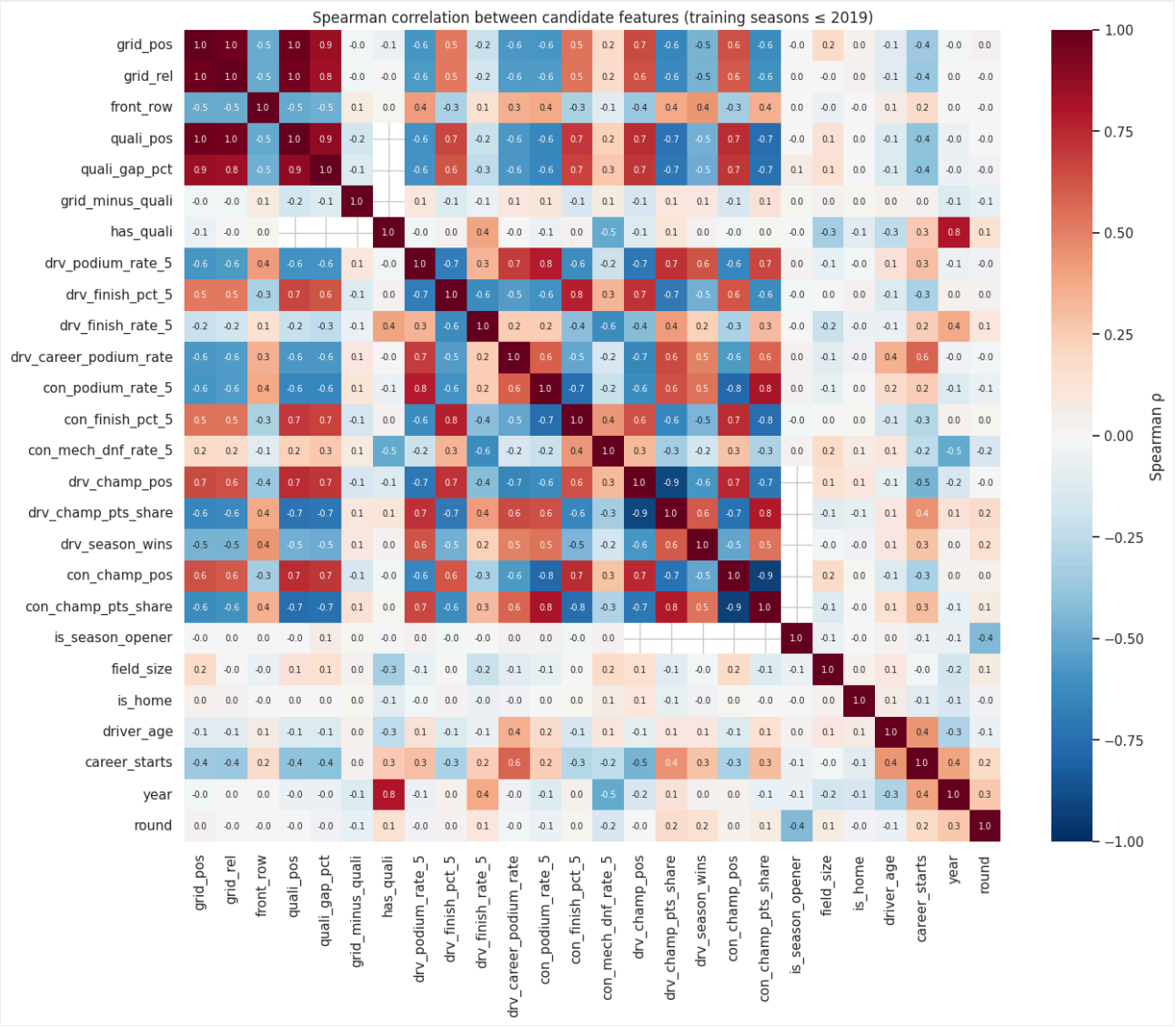


Figure 11. Spearman correlation between candidate features (training seasons). `grid_pos`, `grid_rel` and `quali_pos` are highly correlated (ρ 0.96–0.98), as are each championship rank and its points share (ρ ≈ −0.92); whether to drop one of each pair was tested empirically (Experiment 5.5b).

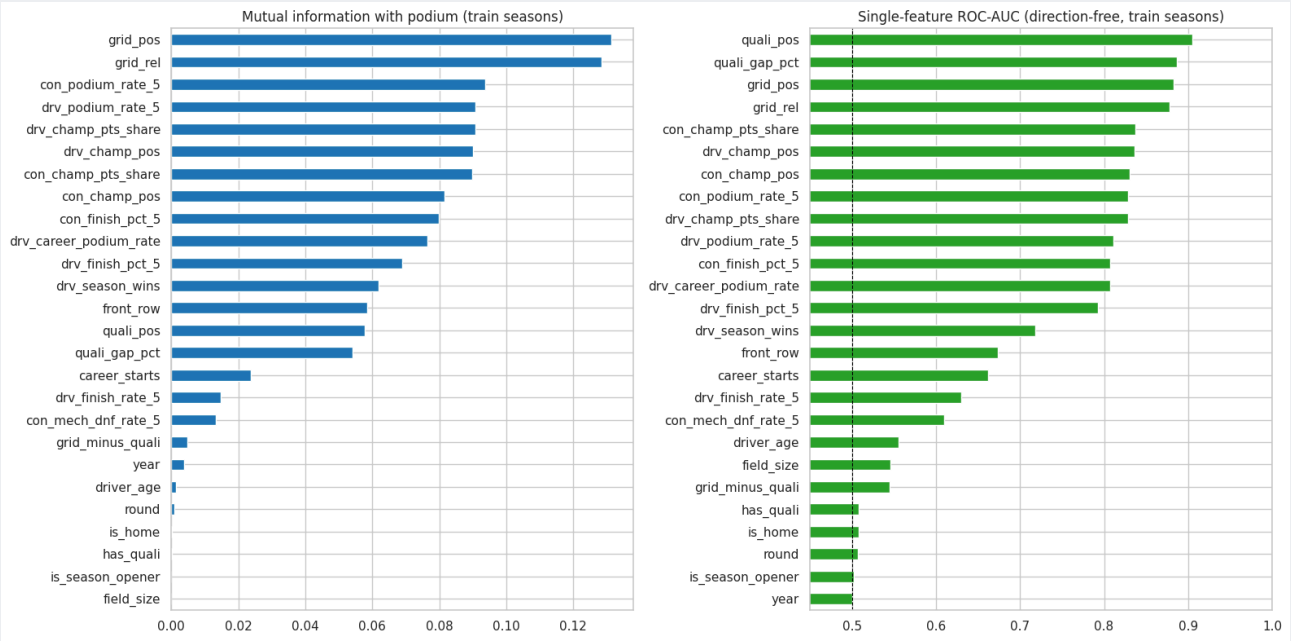


Figure 12. Mutual information with the podium (left) and direction-free single-feature ROC-AUC (right), training seasons. Grid has the highest mutual information (0.13); `year`, `round`, `is_home` and the structural flags are ≈ 0 on their own.

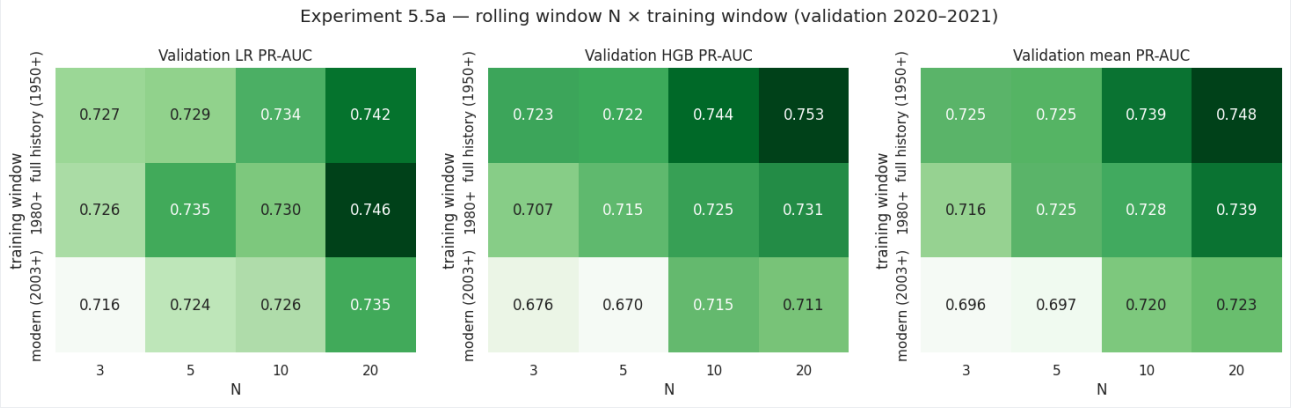


Figure 13. Experiment 5.5a: mean validation PR-AUC of Logistic Regression and HistGradientBoosting for each rolling window N and training window. N = 20 with full-history training (0.748) beats every other combination, so the form features use the last 20 races and training uses 1950–2019.

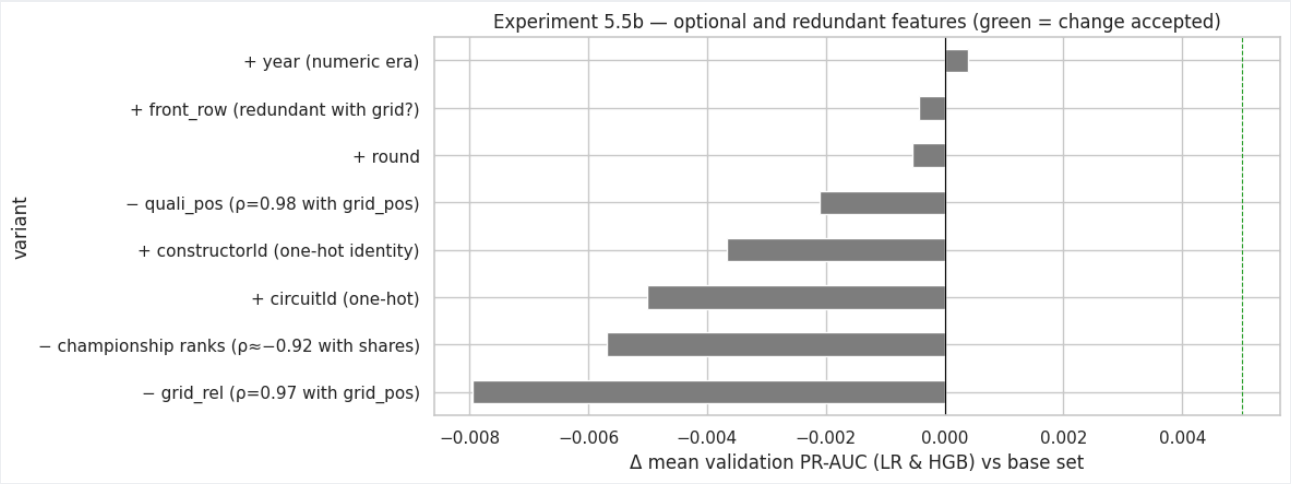


Figure 14. Experiment 5.5b: change in mean validation PR-AUC when an optional feature is added or a redundant one removed (rules fixed in advance: add only if ≥ +0.005; remove only if the score does not drop). No change is accepted. `year`, `round` and circuit/constructor identity do not help, and removing a correlated feature (`grid_rel`, `quali_pos`, championship ranks) lowers the score, so all 24 features are kept.

3. Known Limitations

- **Historical qualifying-coverage drift.** There is no qualifying data before 1994, and it is patchy in 1996–2002, where whole races are missing (e.g. 1 of 17 races in 2001). Coverage is complete only from 2003, so 63% of training rows have imputed qualifying features and the qualifying relationship is learned mainly from 2003+. The `has_quali` flag, the always-available grid features and training-only median imputation mitigate this.
- **Era sensitivity.** F1 changed rules, reliability and field sizes many times. Before 2000 only ≈ 40–50% of entries reached the finish, against 80%+ since 2010, and fields peaked above 30 cars. Early seasons are therefore much noisier: the same logistic model has an in-sample PR-AUC of 0.46 on 1950–79 rows but 0.73 on 2014–19. The test seasons (2022+ ground-effect cars, budget cap) are a new era. Form features adapt within a season, but a new competitive order is only learned after a few races.
- **Small evaluation sets.** 39 validation and 68 test races; 95% race-bootstrap intervals are ±0.06–0.10 PR-AUC, so the four models (test PR-AUC 0.705–0.718) are not meaningfully different. Validation is also used for early stopping, thresholds and model choice, so validation scores are slightly optimistic.
- **Constructor renames reset history.** Team history follows `constructorId`, so Toro Rosso → AlphaTauri → RB starts without constructor form.
- **Judgement calls in cleaning and mapping.** The shared-drive rule (started car, then best `positionOrder`), the non-starter rules, the Indy 500 exclusion and the mechanical/ambiguous status mapping are documented choices; sensitivity checks show the main answers are robust to them.
- **Small home-race sample.** Home starts are ≈ 8% of rows and dominated by British, Italian, French and German drivers, so small home effects cannot be detected.

- **Missing pre-race information.** No weather, tyre strategy, practice pace, upgrades or team orders, and in-race incidents are unpredictable by definition. The most confident test error (Verstappen, 2022 São Paulo GP, predicted 0.88, finished 6th after a collision penalty) is of this kind.
- **Sprint results excluded** for consistency across seasons, at the cost of ignoring Saturday sprint form in 2023–24.
- **Associations, not causes.** The Q1–Q3 answers and the model's feature importances show associations. A feature being important to the model does not mean it causes podium finishes.

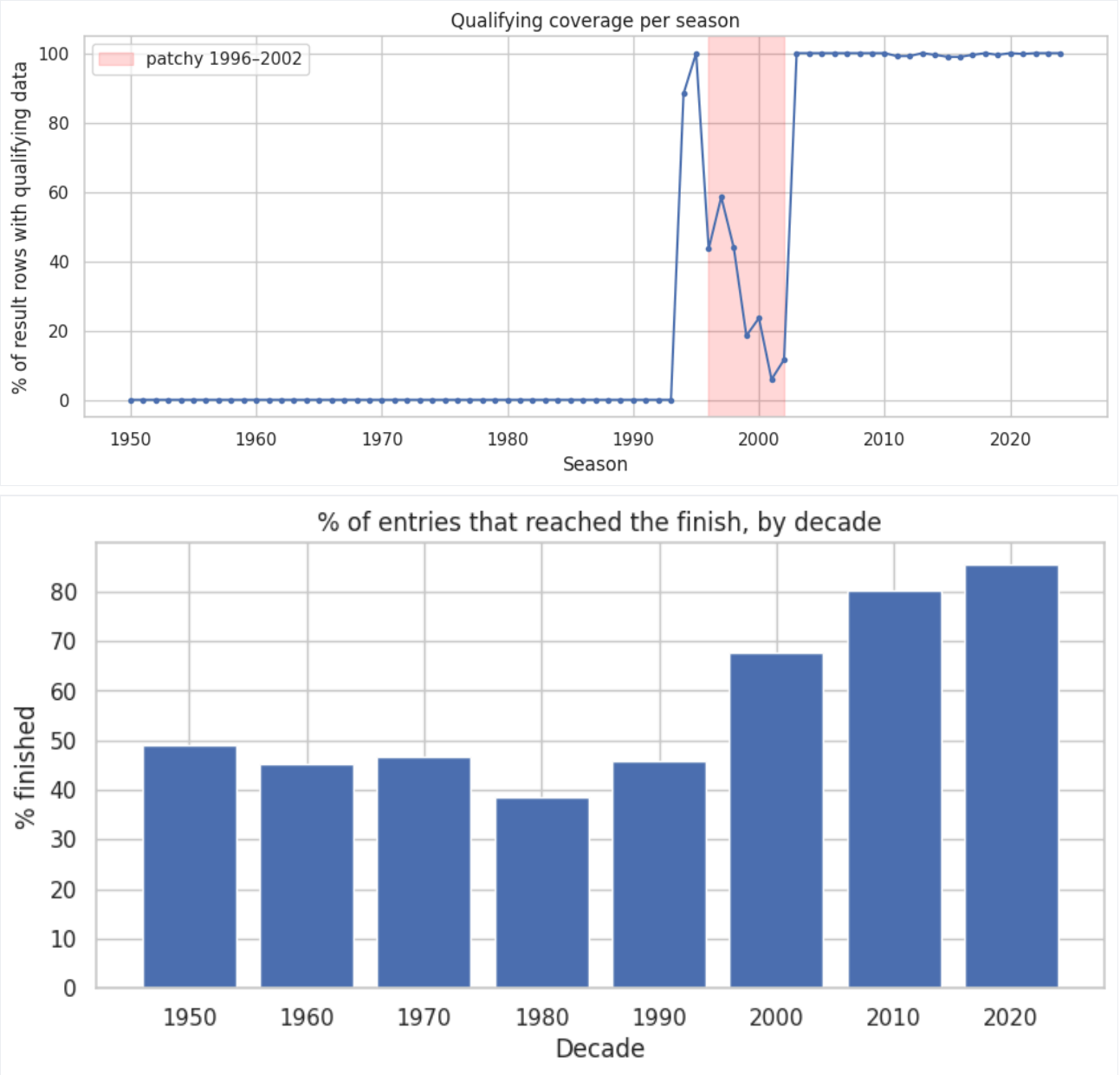


Figure 15. Top: share of result rows with a qualifying record per season, showing the qualifying-coverage drift (red band: patchy 1996–2002). Bottom: share of entries that reached the finish per decade, one source of era sensitivity.